

Assignment

Linear and Polynomial Regression

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Questions

1. You are given the following three data points: (1, 2), (2, 5), and (3, 4). Using the method of least squares, find the equation of the best-fit line in the form $y = wx + b$. Provide the final values for the slope w and the y-intercept b .
2. A data analyst is building a simple linear regression model ($\hat{y} = wx + b$). They plot their data and the resulting best-fit line. Then, they add a single outlier to the dataset that has a high leverage (it's far from the other data points on the x-axis) but is also far from the original regression line. What is the most likely effect of this single new point on the regression line?
 - (a) The slope w and the R-squared value will both likely increase.
 - (b) The slope w will not change, but the R-squared value will decrease.
 - (c) The slope w will change significantly, and the R-squared value will likely decrease.
 - (d) The y-intercept b will change, but the slope w will remain exactly the same.
3. You are trying to model a complex dataset and are testing polynomial regression models with degrees from 1 (linear) to 20. You plot the Mean Squared Error (MSE) on both your training set and your validation set for each degree. Describe the expected shape of the two curves (training error and validation error) on the plot as the polynomial degree increases from 1 to 20. How would you use this plot to choose the optimal degree for your model?
4. **The Mystery Polynomial [Programming Question]:**

You have been given a dataset in `regression_data.csv`. This data was generated from a noisy polynomial function with a degree of 6 or less. Your task is to find this function.

Instructions:

- Load the data and split it into an 80% training set and a 20% testing set.
- Build and evaluate six polynomial regression models with degrees $n = 1$ through $n = 6$.
- For each model, calculate and record error metrics of your choice (e.g., Mean Squared Error, R-squared) on both the training and testing sets.
- Report the most likely polynomial function that generated the data, along with your justification based on the error metrics.

Note: Increasing the degree of the polynomial will obviously reduce training error, so do not rely solely on training errors for model selection.